



RAJALAKSHMI INSTITUTE OF TECHNOLOGY
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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SEMESTER IV

AL23421 - MACHINE LEARNING LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	Weather prediction using decision tree algorithm
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INTRODUCTION

Weather prediction is an important application of Machine Learning that helps in forecasting weather conditions based on historical data. Accurate weather prediction is useful in many fields such as agriculture, transportation, disaster management, tourism, and daily life activities. Predicting weather manually is difficult because weather conditions change frequently and depend on many environmental factors.

Machine Learning techniques can analyze weather data and identify patterns to make predictions effectively. In this mini project, a Weather Prediction System is developed using the Decision Tree algorithm. Decision Tree is a supervised Machine Learning algorithm used for classification and prediction tasks. It works by creating a tree-like structure based on conditions and decisions from the dataset.

PROBLEM STATEMENT

Estimating house prices manually is difficult due to multiple influencing factors such as income level, population density, and geographic location. Hence, an automated system is required to accurately predict house prices using machine learning techniques.

GOAL

The goal of this project is to:

1. To develop a Weather Prediction System using Machine Learning.
2. To predict weather conditions using the Decision Tree algorithm.
3. To train and test the model using a weather dataset.
4. To improve prediction accuracy for weather forecasting applications.

THEORETICAL BACKGROUND

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and make predictions automatically. One of the important applications of Machine Learning is weather prediction, where past weather data is analyzed to forecast future weather conditions. Weather forecasting is useful in agriculture, transportation, aviation, disaster management, and many other fields.

Weather prediction involves analyzing environmental factors such as temperature, humidity, wind, and outlook conditions. Traditional weather forecasting methods require complex calculations and expert analysis. Machine Learning techniques simplify this process by identifying patterns from historical weather datasets and making predictions based on those patterns.

Decision Tree

- ❑ Decision Tree Regression is a Machine Learning algorithm used to predict continuous numerical values.
- ❑ It works by dividing the dataset into smaller parts based on conditions and forming a tree-like structure for prediction.
- ❑ Decision Tree Regression is simple, easy to understand, and useful for predicting values such as temperature, rainfall, and sales data.

Data Set

The weather dataset includes attributes such as Outlook, Temperature, Humidity, and Windy conditions. These features help the model analyze weather patterns and make predictions. Each record in the dataset represents a particular weather condition along with its output value.

The dataset contains two types of outputs:

- Yes – Weather is suitable.
- No – Weather is not suitable.

ALGORITHM EXPLANATION WITH EXAMPLE

Steps:

1. Load housing dataset
2. Preprocess data
3. Split dataset into training and testing sets
4. Train Linear Regression model
5. Train Decision Tree model
6. Predict house prices
7. Evaluate performance

Example:

Input:

Datasets

Weathers(sunny, hot)

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Prediction:

Weather prediction

IMPLEMENTATION CODE AND DATASET USED

➤ Dataset Used – https://github.com/santhoshajith419-lang/ML_miniproject

Category	Description / Example
Dataset Name	weather.csv
Domain	Weather condition
Objective	Classification
Data Source	Dataset
Number of Samples	20
Data Type	Tabular
Instance Description	Income, House Age, Rooms, Population, etc.
Features (Inputs)	Time amount v1 - v28
Feature Types	Text,Datas
Target Variable	weather
Number of Classes	Continuous
License	Open
Version	1.0

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Weather Prediction using Decision Tree Algorithm

Import libraries

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
```

Load dataset

```
df = pd.read_csv("weather.csv")
```

Convert text data into numbers

```
le = LabelEncoder()
```

```
df["Outlook"] = le.fit_transform(df["Outlook"])
df["Temperature"] = le.fit_transform(df["Temperature"])
df["Humidity"] = le.fit_transform(df["Humidity"])
df["Windy"] = le.fit_transform(df["Windy"])
df["Play"] = le.fit_transform(df["Play"])
```

Input and output

```
X = df.drop("Play", axis=1)
```

```
y = df["Play"]
```

Split dataset

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Create Decision Tree model

```
model = DecisionTreeClassifier()
```

Train model

```
model.fit(X_train, y_train)
```

Predict test data

```
y_pred = model.predict(X_test)
```

Accuracy

```
print("Accuracy :", accuracy_score(y_test, y_pred))
```

New weather data

```
new_data = [[2, 1, 0, 1]]
```

Predict result

```
prediction = model.predict(new_data)
```

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```
# Output
print("\n===== WEATHER PREDICTION SYSTEM =====")

if prediction[0] == 1:
    print("Prediction : Weather is Suitable")
else:
    print("Prediction : Weather is Not Suitable")
```

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OUTPUT

```
PS C:\Users\arthi\Desktop\python> & 'c:\Users\arthi\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\arthi\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '62479' '--' 'c:\Users\arthi\Desktop\python\ajith_ml_mp.py'
Accuracy : 1.0
c:\Users\arthi\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
  warnings.warn(

===== WEATHER PREDICTION SYSTEM =====
Prediction : Weather is Not Suitable
PS C:\Users\arthi\Desktop\python>
```

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Model Performance:

The performance of the Weather Prediction System is evaluated using prediction accuracy and testing results. The Decision Tree algorithm provides good performance for classification and prediction tasks because it can easily analyze patterns from the dataset and generate decision rules.

Observations:

- Positive relationship between actual and predicted values
- Some prediction errors exist
- Decision Tree performs better than Linear Regression
- Dataset shows capped values causing clustering

RESULTS AND FUTURE ENHANCEMENT

Results

- Model successfully predicts house prices
- Linear Regression provides baseline performance
- Decision Tree improves prediction accuracy
- System works efficiently for prediction tasks

Future Enhancements

- Use advanced models like Random Forest or Gradient Boosting
- Improve feature selection
- Develop web-based application
- Use real-time real estate data

Git Hub Link of the project along with the report	https://github.com/ajithkumar-sys/ML-PROJECT
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REFERENCES

Ethem Alpaydın – Introduction to Machine Learning

- Scikit-learn Documentation
- California Housing Dataset
- Python Documentation